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# Heat Stroke Burden and Validity of Wearable-Derived Core Temperature Estimation during Elite Military Training

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Title Page

**Manuscript Title:** Heat Stroke Burden and Validity of Wearable-Derived Core Temperature Estimation during Elite Military Training

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**Abstract**

**Objectives**: Exertional heat stroke (EHS) remains a significant health concern while training in hot environments, prompting the development of non-invasive wearable technology for monitoring estimated core temperature (ECT). The objective of this study was to review the effectiveness of an ECT device during elite military training by providing a case series of individuals who developed EHS.

**Methods**: This was a retrospective study of EHS cases during high-intensity training while wearing the ECT device, Zephyr Bioharness™. Data was collected from January 2021 through September 2021 at the Air Force Special Warfare Training pipeline in San Antonio, TX. Rectal temperatures of EHS diagnoses, defined by central nervous system (CNS) dysfunction and rectal temperature approaching or >40 deg C (104 deg F), were compared to ECT reading via Zephyr Bioharness™. Incidence rates and psychometric properties were calculated using R package.

**Results:** 47,058 daily peak ECT measurements were collected among 1,364 trainees. 499 trainees flagged as potential EHS by Zephyr Bioharness™ reading >39.7 °C (103.5°F). Incidence of confirmed EHS was 0.8/ 1000 person-months. Of the 10 confirmed EHS cases (9 males, 1 female; age=23.4 ± 2.7 yrs; BMI=25 ± 2; body fat=13 ± 5%), 8 trainees had a peak ECT reading below 39.7 °C which resulted in a sensitivity of 20%, specificity of 98.9%, positive likelihood ratio (LR) of 18.93 (95% Confidence Interval[CI] 5.5-65.6) and a negative LR of of 0.81 (95% CI 0.6-1.1).

**Conclusion**: ECT had substantial false positive and negative rates. Further studies are needed to validate this technology in other populations, and the algorithm used in this device needs to be

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refined to better capture the environmental and physical requirements in the special operations population.

**Key words**: heat illness; sports medicine; Air Force; Special Operations; military medicine; heat safety

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## Introduction

Exertional heat illness (EHI) remains a significant cause of morbidity and mortality in the athletic population, and an occupational hazard to military personnel who are often tasked to perform training and operations during summer months [1–4]. EHI encompasses a range of injury severity from the mildest form, exertional heat cramps , to a life-threatening condition, exertional heat stroke (EHS) [3,5] EHS is defined as central nervous system (CNS) dysfunction with hyperthermia, typically a core body temperature greater than 40$^{o}$C (104$^{o}$F) [2,4,6,7]. CNS dysfunction may present as abnormal gait, confusion, agitation, seizures, or syncope, and is associated with heat-induced thermoregulatory failure of the hypothalamus [8]. According to the American College of Sports Medicine (ACSM) consensus statement released in 2021, even with the multitude of educational and preventive efforts across a variety of platforms, there appears to be no change in the number of EHS deaths in the athletic populations [2].

Untreated EHS has nearly an 80% mortality [9]. The gold standard treatment of EHS is early recognition and aggressive cooling to provide rapid reversal of hyperthermia [2,6,7,10]. If treated appropriately, mortality is zero [11]. If not promptly and properly treated, EHS can result in serious long-term complications from the release of cytokines that induce a massive systemic response resulting in coagulopathies that can lead to disseminated intravascular coagulation, and ultimately multi-organ failure [1]. Recent evidence also suggests possible long-term

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cardiovascular complications and increased incidence of atrial fibrillation post EHS, though the mechanism of action remains unclear [2].

The incidence of EHS appears to be increasing globally, and will likely continue to increase due to the rise of global temperatures and increasing number of endurance athletes and events, making rapid recognition and treatment even more important [2,12]. A five-fold increase in hospitalizations was reported from EHS in the United States military between 1980 and 2001 [1]. As such, there is growing interest among multiple stakeholders for the development of wearable technology to accurately estimate core temperature and thus mitigate the risk of EHS. Although rectal temperature is best practice for measuring core temperature in the diagnosis of EHS [7], it is not practical to use during physical activity. Another device that has been extensively studied among the sports population is ingestible gastro-intestinal (GI) capsules, which have been found to be comparable in accuracy to rectal temperature in measuring core temperature [13]. Unfortunately cost and practical limitations prevent the use of this device in mass populations.

The Zephyr Bioharness™ (ZB; Medtronics [Boulder CO]) is a wireless chest strap that purportedly provides real-time estimates of core temperature from a proprietary algorithm based on the wearer's heart rate (HR)[14]. The <<institution>>, housed at <<location>>, first employed this device in 2019 as a means to monitor trainees' estimated core temperature during activity due to the increased risk of EHS caused by the hot climate of the area and physical demands of the training program. However, no studies have been performed to observe the effectiveness of Zephyr Bioharness™ in detecting exertional heat illnesses, primarily the life-threatening form of EHS. The purpose of this paper is to evaluate the ability of Zephyr Bioharness™ in identifying individuals at risk of developing EHS during physical activity in military training.

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## Methods

*Design and Setting*

The <<institution>> Institutional Review Board approved this study protocol. This study was conducted at <<location>>. From January 2021 through September 2021, 1,364 trainees were tracked during all outdoor physical training, and ECT was continuously monitored in real time during all strength and conditioning, running, and rucking activities while wearing physical training (athletic attire) and utility uniforms (with and without blouse, body armor, or helmet). Peak ECT was recorded for each individual session, and environmental conditions were measured by local wet bulb globe temperature (WBGT) devices (Kestrel Meter 5400 Heat Stress Tracker).

Trainees wore the Zephyr Bioharness™ device during physical activity and their ECT was monitored by trained staffon a tablet screen display. When a trainee's ECT approached 39.7 °C (103.5 °F) on the tablet, the trainee was closely monitored for previously established signs and symptoms of EHS. If ECT reached 39.7 °C (103.5 °F), the trainee stopped training and was asked by trained medical staff how they were feeling. If there were no apparent signs or symptoms of CNS dysfunction, the trainee was allowed to continue training but closely monitored. In the event of CNS dysfunction, the trainee was immediately assessed with a rectal core temperature using the DataTherm II device, and inserting the rectal probe to the depth of 15 centimeters (6 inches). If EHS was confirmed, the individual was treated by medical staff with cold water immersion. Participant demographics are summarized in Table 1. Electronic medical records were then retrieved to identify and review EHS cases during that time period, and the cases were confirmed with an internal tracker utilized by the <<institution>> medical staff. EHS was diagnosed by an on-site physician and defined as CNS dysfunction with hyperthermia,

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although a core temperature >40 °C (104 °F) was not required for the life-saving treatment of rapid cooling to be initiated [6].

*Statistical Analyses*

Demographic information was recorded as mean ± standard deviation (SD). The incidence of confirmed EHS,proportion measures of predicted to confirmed EHS cases, psychometric properties and figures were calculated using R Version 4.0[15]. Additionally, a Bland-Altman test was performed to test limits of agreement between ECT and rectal temperature measurement for the 10 confirmed cases.

**Results**

47,058 daily peak ECT measurements amongst 1,364 trainees were collected during the study epoch. The mean ambient temperature was 26.7 ± 12.2°C (80.1 ± 54.0°F) with a 72 ± 15% relative humidity. There were a total of 499 flagged ECT measurements (1.1%) as potential EHS based on estimated core temperature reading via Zephyr Bioharness™ >39.7°C ( 103.5 °F). Of those flagged, only 2 trainees had confirmed EHS assessed using rectal temperature and CNS assessment. Incidence of confirmed EHS was 0.8 per 1000 person-months. See Figure 1.

A total of 10 athletes were diagnosed with EHS during the study period. Of the 10 confirmed EHS cases (100% males; age=23.4 ± 2.7 yrs; BMI=25 ± 2; body fat=13 ± 5%), 8 (80%) had a predicted peak ECT reading below the threshold of 39.7 °C. See Table 2. This resulted in a sensitivity of 20%, specificity of 98.9%, positive likelihood ratio (LR) of 18.93 (95% Confidence Interval[CI] 5.5-65.6) and a negative LR of 0.81 (95% CI 0.6-1.1). A Bland-Altman plot is shown in Figure 2, demonstrating low agreement between the two core temperature measurements.

**Discussion**

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The primary purpose of this paper was to evaluate the effectiveness of Zephyr Bioharness™ in identifying individuals at risk of developing EHS. Based on our results, there are several concerns regarding the validity of ECT from a wearable device. The biggest concern is the lack of sensitivity in detecting development of EHS. Of the 10 trainees who were diagnosed with EHS by a physician, only two were identified above the defined temperature threshold by the device. Given the morbidity and mortality associated with EHS and the poor sensitivity of this device (20%), extreme caution is warranted when using this technology to monitor individuals at risk for developing EHS while exercising in the heat.

Another concern with this wearable device is the poor specificity/poor positive predictive value. In this study, 499 candidates were flagged as potential EHS, while only two of those developed EHS. Requiring medical staff to review flagged individuals takes a significant amount of time and resources, which could be more appropriately used to monitor a large group of individuals and detect early clinical signs and symptoms. Finally, the algorithm calculates the ECT based on heart rate and connection quality, or "confidence," of the device reading. When confidence is at 100%, there is approximately a ±1° C (±2° F) range of the true value of core temperature [16]. However, the confidence dramatically decreases when a trainee sweats, exercises in uniforms that may interfere with the device, or when doing activities that may cause the device to move or rub. Not only is unnecessary manpower required to confirm that a suspected heat casualty (via Zephyr Bioharness™) is cleared to continue training, but the majority of those who suffer a heat illness never exceeded the 39.7 °C (103.5 °F) threshold on the Zephyr Bioharness™ device. To highlight this point, Figure 1 represents peak Zephyr Bioharness™ ECT readings during fitness days between January through September 2021 for trainees. Note how several dozen encounters show trainees well above the 40 °C (104 °F)

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concern for a heat illness, yet the vast majority of those who eventually succumbed to an EHS never approached the 103.5 ° F threshold set by the Zephyr Bioharness™ device. The wide spread of ECT throughout data collection is particularly alarming because it drastically over- and under-predicted core temperature.

To our knowledge, the Zephyr Bioharness™ has never been compared to a validated device for monitoring core temperature in the setting of an exertional heat stroke. While the Zephyr Bioharness™ is marketed for early detection of EHS, available literature offers data only on the psychometric properties of Zephyr Bioharness™ (recording heart rate and respiratory rate), and the studies were not used to validate the device for detecting or mitigating the risk of EHS [14,16,17]. The participant population of each study was small thereby decreasing the power of the studies. Additionally, there was some discordance among the data in these studies [14]. Moreover, the data was in relation to low to moderate intensity exercises.

Literature on the Zephyr Bioharness™ has indicated that ECT tends to be more accurate on individuals with higher body fat percentage as compared to those with lower body fat percentage [16]. The results may be due to differences in surface area and metabolic demand, but this is important for the purpose of this study as most elite military trainees have lower body fat percentages. While the studies reviewed by Nazari et al (2018) may have demonstrated good to excellent overall evidence in favor of reliability and validity of Zephyr Bioharness™ to report RR and HR values [14], our study shows poor sensitivity and specificity for Zephyr Bioharness™ to be used for ECT monitoring. Based on our results, we recommend that the device algorithm for ECT should be improved and tailored to specific populations, such as in military personnel with low body fat and prolonged exertion while wearing a considerable amount of equipment.

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There are other wearable devices that have been studied that gather additional data as an attempt to improve on the current algorithms. One such device was reviewed and published in January 2022 by Buller et al. Their team investigated HR-based algorithms for ECT using a chest-worn sensor, combined with gait instability from a trunk-worn sensor to predict EHS [18]. The study was done using US military personnel during long runs and road marches while wearing heavy military gear. Among 1806 participants, the six who experienced EHS were identified as having elevated core temperature and/ or ataxic gait by the combined algorithms. While there were no reported false-negatives, there were 209 total false positives in which both faculty and participants did not observe or experience, respectively, exertional heat illness symptoms. While this study is promising, we will need additional data before adopting this technology to large-scale use in the military and athlete population.

Early recognition and rapid treatment of EHS is critical in prevention of death or significant morbidity of athletes, workers and military members. Thus, there is significant demand for development of a non-invasive device to detect early signs of EHS and ultimately prevent EHS. The role of non-invasive wearable devices for body physiology monitoring is hypothetically attractive, however remains controversial and requires further analysis prior to real-time use to ensure the safety of all participants.

In summary, using a tool with poor sensitivity and specificity that currently has, at the very least, ±1 °C (±2 °F) range between a reading and true core temperature, is not helpful and potentially dangerous. Additionally, although not accounted for in this study, errors may be even further compounded with body composition, movement speed, and hydration status. Appropriate management of EHS could potentially be delayed with too much emphasis placed on a device that has suboptimal data for reliability and validity.

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## Conclusion

Early recognition of potential EHS relies on the reliability and validity of the device being used and at present, the predicted ECT from the Zephyr Bioharness™ device has poor sensitivity and poor negative predictive value based on our findings. Extreme caution is warranted when using this technology to monitor individuals at risk for developing EHS while exercising in the heat, and we recommend the clinicians and medical teams on the ground be the first line of defense in detecting EHS. Further studies are needed to validate improvements upon technology within the military and athletic populations.

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**Figure 1 Legend:**

**Figure 1. Fire Plot of Maximum Zephyr Bioharness™ ECT of All Trainees.** Color dots represent daily peak ECT measurements of 1,364 military trainees, and the dotted line represents the level at which the Zephyr Bioharness™ device flags temperature over 103.5 °F. A total of 499 trainees were flagged as potential EHS. Confirmed heat strokes are labeled with black dots.

Of the 10 reported heat strokes, only two had a maximum Zephyr Bioharness™ ECT above 103.5 °F.

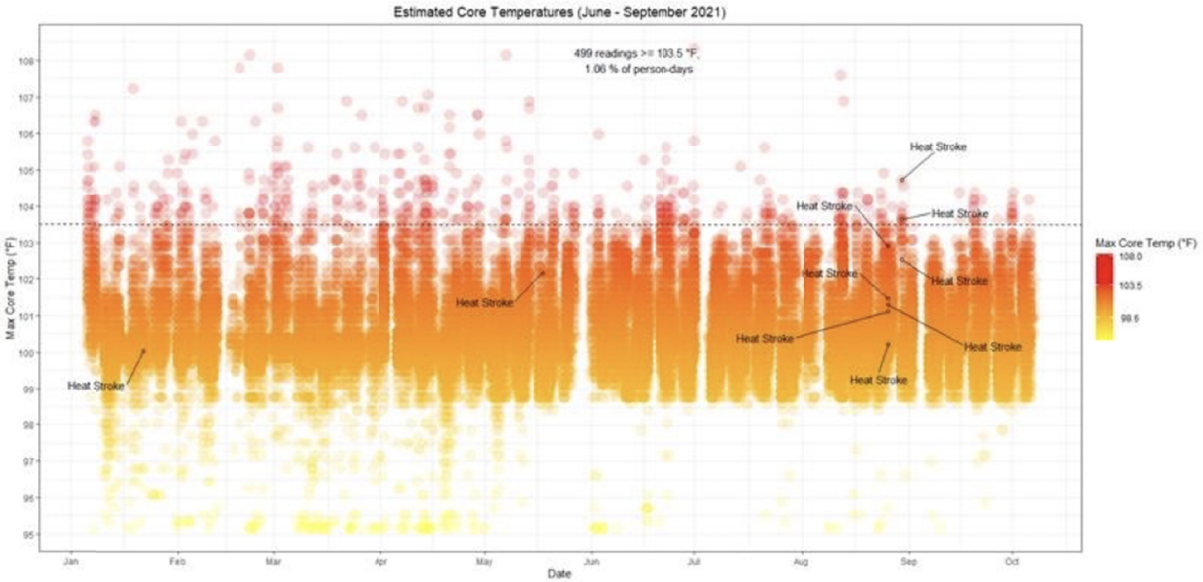


**Figure 1. Maximum Zephyr Bioharness™ ECT of All Trainees. Of the 10 reported heat strokes, only two had a maximum Zephyr ECT above 103.5 °F**

**Figure 2 Legend:**

**Figure 2. Bland-Altman Plot to test line of agreement between rectal temperature and estimated core temperature with Zephyr device.** Red lines represent 95% confidence intervals.

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**Figure 2. Bland-Altman Plot to test line of agreement between rectal temperature and estimated core temperature with Zephyr device.**

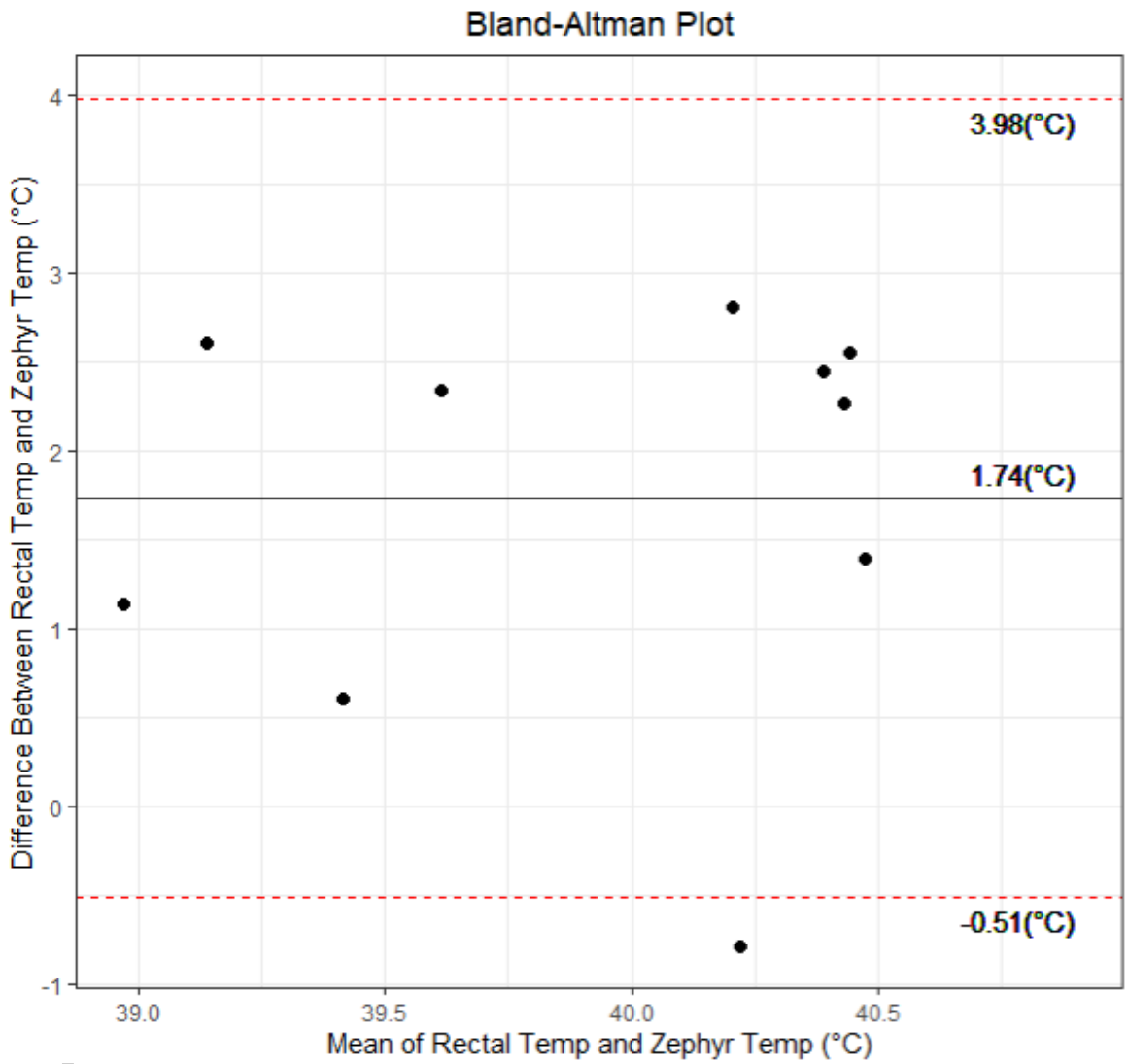

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**Table 1.** Participant demographics (mean ± SD).

| N | Age (yrs) | Sex (M/F) | Height (cm) | Weight (kg) | LBM (kg) | Body Fat % | BFM (kg) | BMI |
|---|---|---|---|---|---|---|---|---|
| 1,364 | 23.6 ± 4.0 | 1,351 M<br>12 F | 177 ± 6.6 | 78.5 ± 8.4 | 69.4 ± 7.6 | 11.8 ± 3.4 | 9.3 ± 2.9 | 25.0 ± 2.1 |

LBM (lean body mass); BFM (body fat mass); BMI (body mass index)

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Table 2

| Age (yrs) and Gender (M=male, F=female) | Height (cm) | Weight (kg) | Event leading up to EHS | WBGT at time of incident (Celcius) | Signs/Symptoms | Peak rectal temp (Celcius [Fahrenheit]) | Peak Zephyr ECT (Celcius [Fahrenheit]) | Treatment | Outcome |
|---|---|---|---|---|---|---|---|---|---|
| 19, M | 177.8 | 72.4 | 3-mile run | 13.9 | Bilateral arm numbness followed by bilateral leg numbness with brief LOC | 40.4 (104.8) | 37.8 (100.1) | CWI within 5 minutes of collapse | Gradual RTT |
| 23, M | 190.5 | 97.5 | 4-hour run in full kit with helmet | 14.4 | Fell, hit jaw on rocks with subsequent LOC | 41.6 (102.5) | 39.2 (102.5) | CWI within 7 minutes of symptoms | Gradual RTT |
| 24, M | 175.3 | 88.0 | Time trial rucking followed by run in full kit | 20.7 | Altered mental status | 41.7 (107.1) | 39.2 (102.5) | CWI within 5 minutes of symptoms | Gradual RTT |
| 24, M | 190.5 | 95.3 | 1.5 hrs exertion in full kit, cals, followed by 2-mile ruck | 21.0 | Collapsed during ruck with brief LOC | 39.8 (103.7) | 38.8 (101.8) | CWI within 9 minutes of symptoms | Gradual RTT |
| *23, M | 190.5 | 97.5 | Extended duration ruck march | 29.9 | Collapsed during final assessment | 42.2 (107.9) | 40.6 (105.1) | CWI | Gradual RTT |

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| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| 30, M | 188.0 | 97.5 | 6-hour ruck/run and pits workout | 23.9 | Fatigue, poor performance, difficulty standing | 41.6 (106.8) | 39.3 (102.7) | CWI within 2 minutes of symptoms | Gradual RTT |
| 22, M | 182.9 | 70.3 | Extended duration ruck march | 25.0 | Dizzy, altered mental status | 39.4 (103) | 38.4 (101.1) | CWI | Gradual RTT |
| 22, F | 188.0 | 85.3 | Extended duration ruck march | 25.0 | Altered mental status | 40.8 (105.4) | 38.4 (101.2) | CWI within 10 minutes from symptoms | Gradual RTT |
| 26, M | 182.9 | 86.2 | 8-mile ruck | 23.9 | Bilateral leg/arm weakness, fatigue | 41.2 (106.1) | 39.8 (103.6) | CWI | Gradual RTT |
| 23, M | 175.3 | 72.6 | Ruck and mud pit calisthenics | 23.9 | Unsteady gait, altered mental status | 39.6 (103.2) | 39.0 (102.2) | CWI within 11 minutes | Gradual RTT |

Cm (centimeters); kg (kilograms); EHS (exertional heat stroke); WBGT (wet bulb globe temperature); CWI (cold water immersion); RTT (return to training); LOC (loss of consciousness)
* (recurrence of episode with same trainee)

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